

Decomposing Rookie Potential in NBA with a Mixture-of-Factors Model

Chuyang Su, Kangyu Zhao
12/03/2025

Data Description and Preprocessing

Feature type:

- Numeric (counts, rates)
- Categorical labels
- Advanced metrics

Dropped 246 players with no games played

Final dataset size: 1616 players

Dropped 7 players with no games played

Final dataset size: 53 players

Box Plus/Minus Regression		
Variable	Coefficient	
	Position 1.0 (PG)	Position 5.0 (C)
Pts (adjusted for team context)	0.860	
3PM (3 pointer made)	0.389	
AST	0.580	1.034
TO	-0.964	
ORB	0.613	0.181
DRB	0.116	0.181
STL	1.369	1.008
BLK	1.327	0.703
PF	-0.367	
Variable	Coefficient	
	Offensive Role 1.0 (Creator)	Offensive Role 5.0 (Receiver)
FGA	-0.560	-0.780
FTA (=0.44*FGA)	-0.246	-0.343

So, to calculate **VORP**, the formula is simply: $[BPM - (-2.0)] * (\% \text{ of possessions played}) * (\text{team games}/82)$.

WIN SHARES PER 48 MINUTES

$$WS/48 = 48 \times (\text{Win Shares} / \text{Minutes Played})$$

MARGINAL OFFENSE

$$(\text{points produced}) - 0.92 \times (\text{league points per possession}) \times (\text{offensive possessions})$$

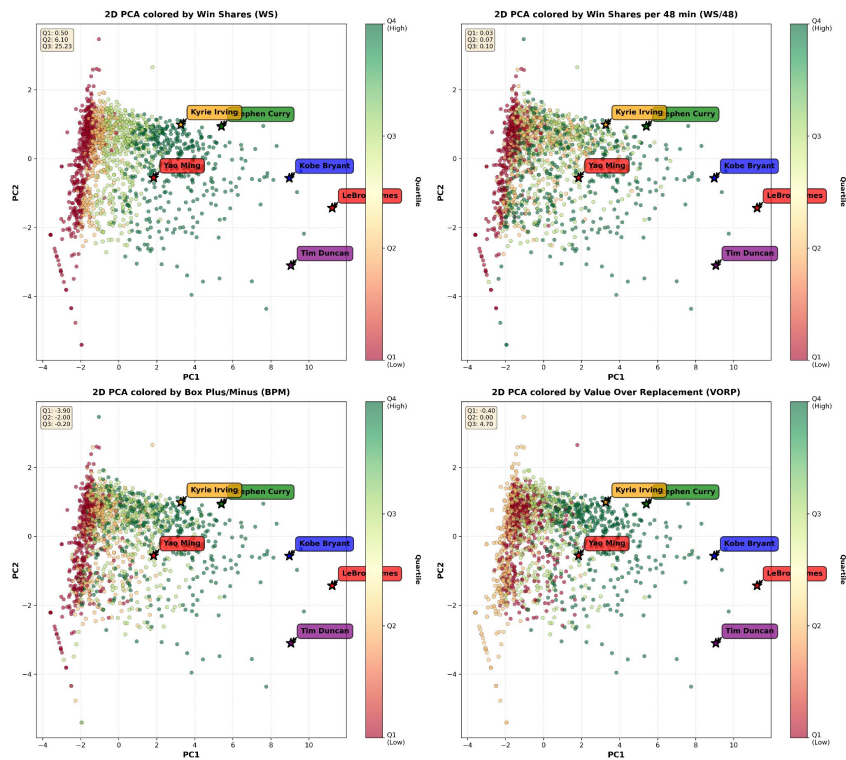
MARGINAL POINTS PER WIN

$$0.32 \times (\text{league points per game}) \times ((\text{team pace}) / (\text{league pace}))$$

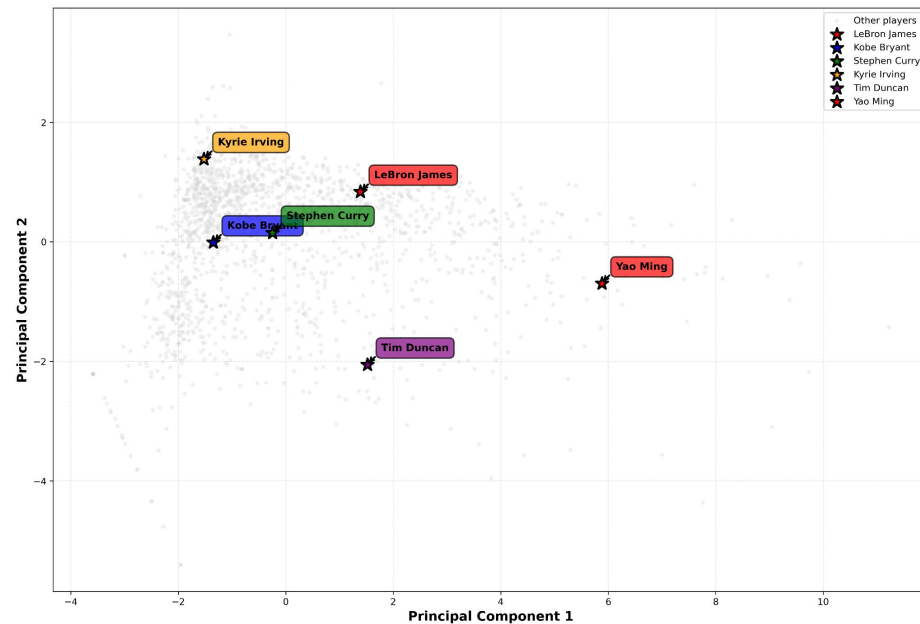
OFFENSIVE WIN SHARE

$$(\text{marginal offense}) / (\text{marginal points per win})$$

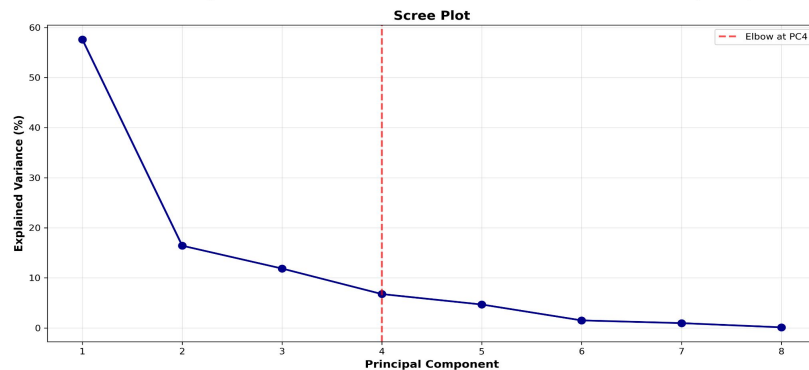
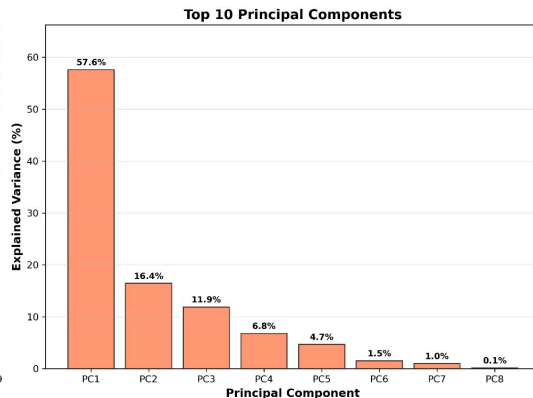
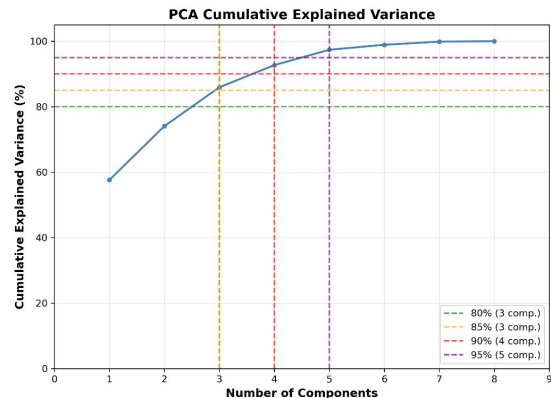
Exploratory Data Analysis - PCA



2D PCA: NBA Draft Players with Star Highlights



Exploratory Data Analysis - PCA



=====

PC1 (explains 57.62% variance)

=====

Top 10 features:

1. minutes_played	+ 0.4583
2. points	+ 0.4473
3. games	+ 0.4431
4. total_rebounds	+ 0.4030
5. assists	+ 0.3907
6. free_throw_percentage	+ 0.1806
7. 3_point_percentage	+ 0.1648
8. field_goal_percentage	+ 0.1369

=====

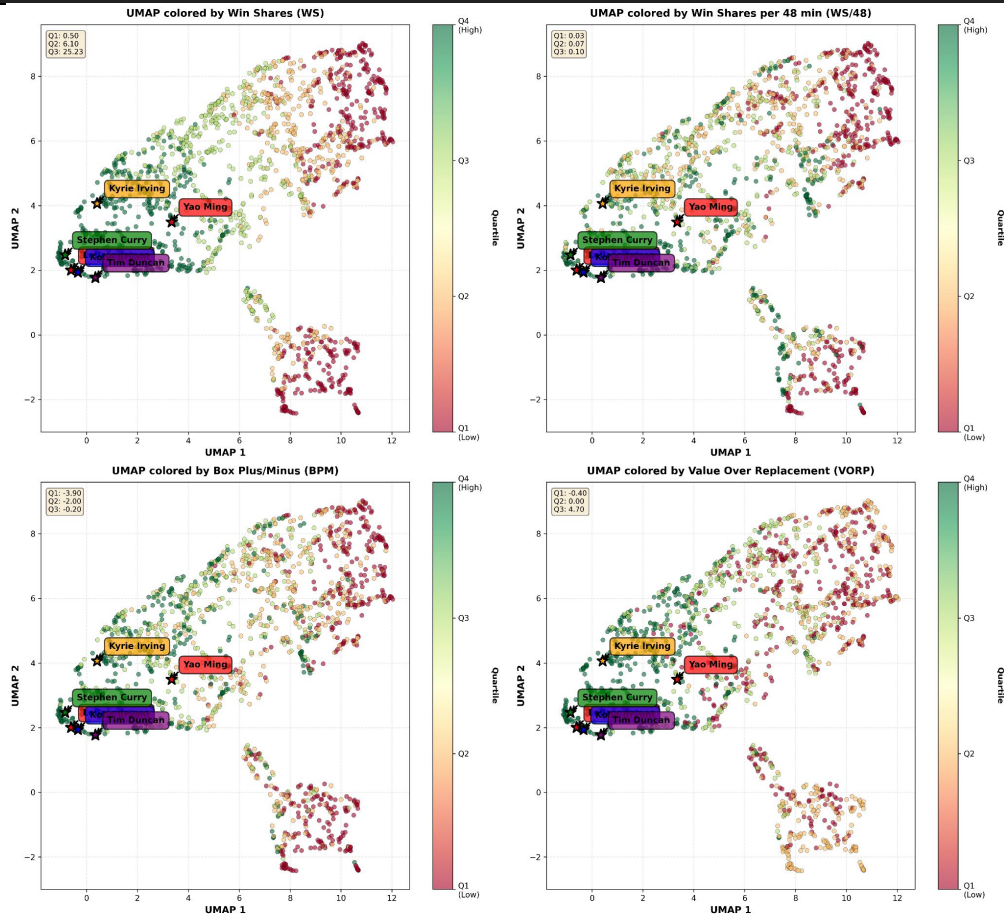
PC2 (explains 16.43% variance)

=====

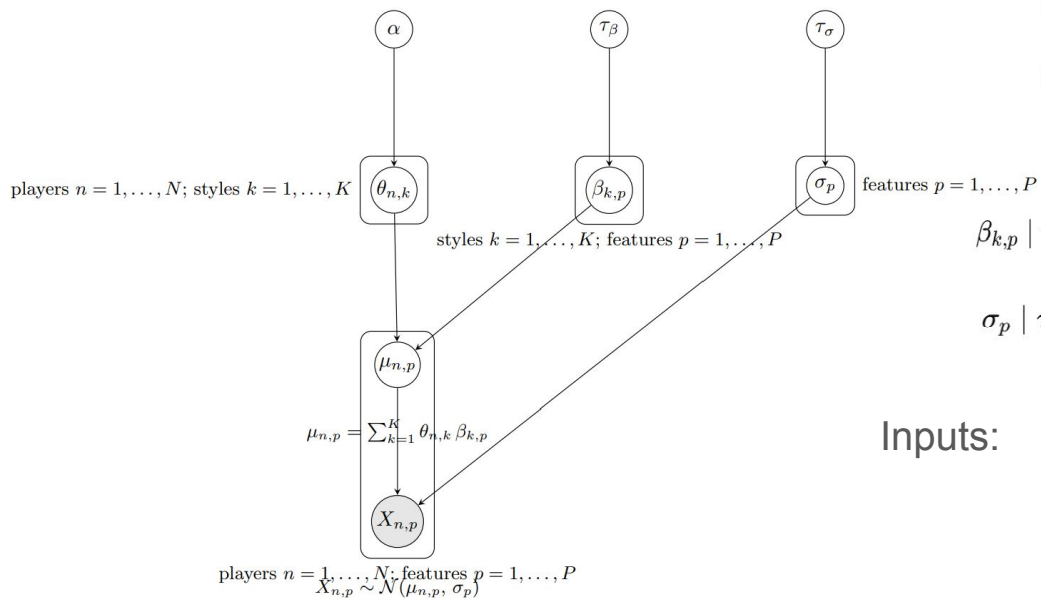
Top 10 features:

1. 3_point_percentage	+ 0.6765
2. free_throw_percentage	+ 0.6155
3. total_rebounds	- 0.2715
4. field_goal_percentage	- 0.2679
5. games	- 0.0799
6. minutes_played	- 0.0775
7. points	- 0.0565
8. assists	+ 0.0503

Exploratory Data Analysis - UMAP



Model Setup



$\theta_n = (\theta_{n,1}, \dots, \theta_{n,K})$ lives on the K -simplex.

$\theta_n \mid \alpha \sim \text{Dirichlet}(\alpha \mathbf{1}_K), \quad n = 1, \dots, N,$

$\beta_{k,p} \mid \tau_\beta \sim \mathcal{N}(0, \tau_\beta^2), \quad k = 1, \dots, K, \quad p = 1, \dots, P.$

$\sigma_p \mid \tau_\sigma \sim \text{Half-Cauchy}(0, \tau_\sigma), \quad p = 1, \dots, P.$

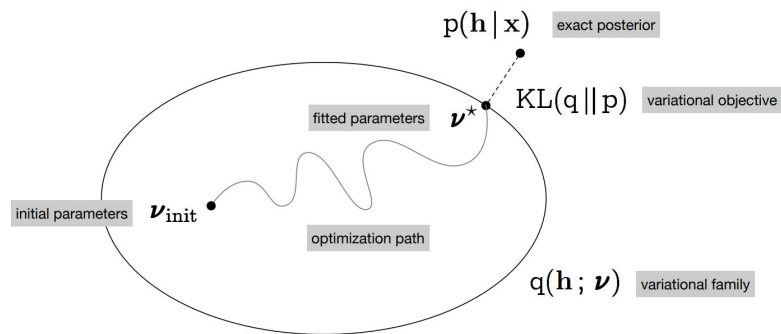
Inputs:

"PTS per 36",
 "REB per 36",
 "AST per 36",
 "Avg minutes",
 "FG% (logit)",
 "3P% (logit)",
 "FT% (logit)",
 "WS/48",
 "BPM",
 "VORP",

Outputs:

$p(\theta, \beta, \sigma, \mu \mid X)$

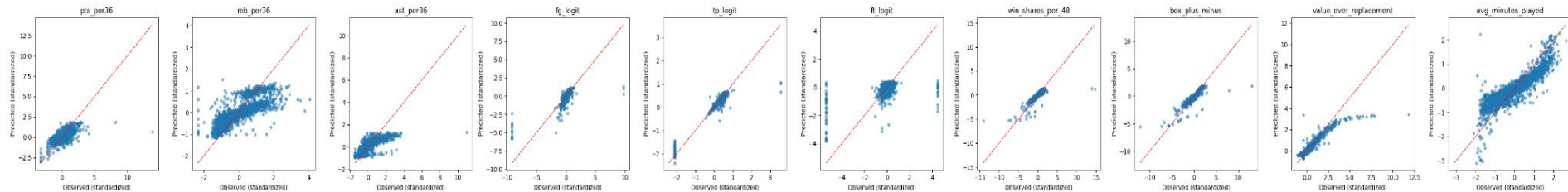
Model Training and Evaluation



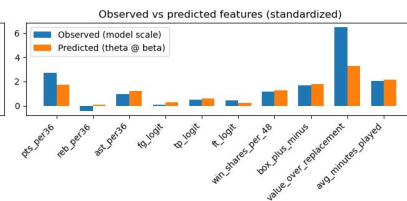
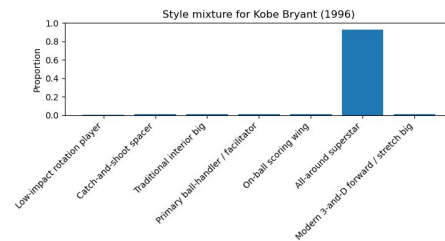
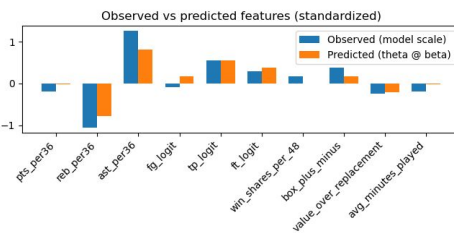
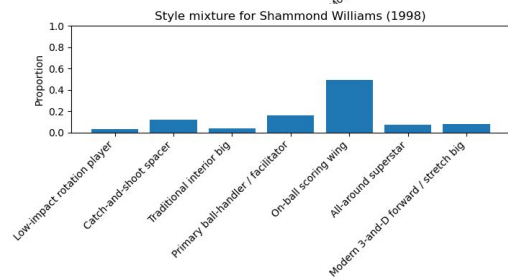
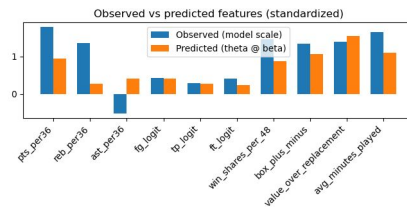
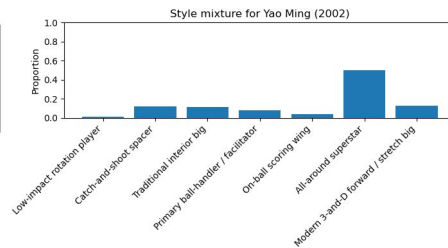
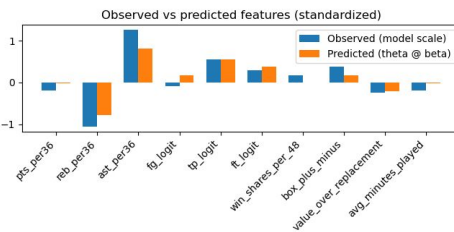
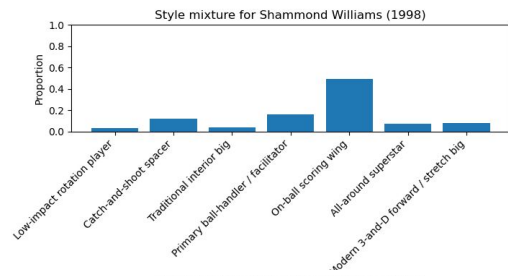
Cross-validation Results for Gaussian Style-Topic Model

K	alpha	Mean MSE	Std MSE	Rank
3	0.5	1.0778	0.0884	2
5	0.5	1.1330	0.0487	3
7	0.5	1.0741	0.0314	1

Reconstruction Loss:

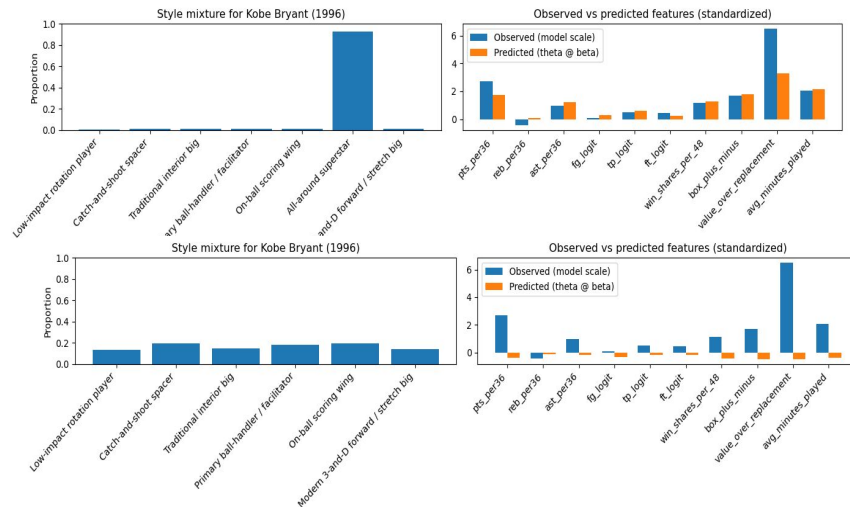


Domain Performance of Selected Model



To be continued: Idea and Challenge

- Degenerate to trivial clustering outcomes, which does not show different patterns between superstars.
- Try to avoid collapse by dropping the most explained principle component.



For Project details, please refer to our github repository:

<https://github.com/Schuyn/Unsupervised-Machine-Learning-Final-Project>